# Missing-Data Sensitivity of the INCOME Effect

A re-analysis of Metrick and Schmelzing (2024, NBER w29281)

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**Abstract**

We re-analyze the regressions in Metrick and Schmelzing [2024] under explicit treatment of the missing-not-at-random (MNAR) selection that drops 638 of 910 crises (70%) from the complete-case sample. A layered approach—selection models, copula sensitivity, profile likelihood, MNAR multiple imputation, and Horowitz–Manski worst-case bounds—produces three findings. First, the paper's continuous-outcome INCOME coefficient on the post-crisis GDP gap is robust across eight independent specifications: complete-case OLS ($-0.246$), Heckman maximum-likelihood ($-0.235$), MICE-MNAR pooled ($-0.235$), and GJRM cross-copula range 0.021 all agree to within 5%. Second, the binary intervention-dummy regressions in Table 2 are imputation-assumption-dependent: MICE-MNAR shifts the log-odds INCOME estimate by 10–78% from complete-case in five of six interpretable outcomes, and the no-assumption Horowitz–Manski bounds contain zero on all six. Third, the methodological contribution is the layered diagnostic: each of the four lenses isolates a different risk to identification, and only their joint application discriminates the robust headline (continuous outcome) from the precision-overstated one (binary outcomes).

## 1 Problem

Metrick and Schmelzing [2024] regress crisis-response outcomes on country-time covariates that include log lagged GDP per capita ("INCOME"), Polity, the Ilzetzki–Reinhart–Rogoff fine FX-regime classification, public-debt-to-GDP, and government expenditure to GDP. The headline regressions cover an asserted historical span of 33 to 2019, but Tables 2 and 3 are estimated complete-case: any crisis with one or more missing covariates is dropped. The complete-case subsample has $N = 272$ in our pipeline and $N = 273$ in the paper.[1]

The dropped subsample is not random. Median year among kept crises is 1996.5; among dropped, 1900 overall and 1916 post-1800. The paper's quantitative claims therefore speak predominantly to the post-Bretton-Woods period despite the 33–2019 narrative span. Missingness is driven by the historical availability of structured macro-statistical series: the FX-regime variable comes from Ilzetzki–Reinhart–Rogoff (starts 1940), and the fiscal pair from Mauro et al. (sparse pre-1970). Table 1 reports per-regressor missingness in the post-1800 sample. By era, almost every dropped crisis is pre-1946 (107 vs 0 kept in 1800–69; 125 vs 0 in 1870–1913; 158 vs 1 in 1914–45). Selection on observables alone cannot fix this—missingness is structurally tied to historical period and country size, both correlated with crisis outcomes.

*All numerical claims in this memo are sourced from the analysis pipeline in `scripts/00_etl.R`–`scripts/05_sensitivity.R` and the rendered tables under `output/tables/` (referenced below by filename only). Replication code and data-construction notes are in the project's `README.md` and `INSTALL_NOTES.md`.

[1] The $-1$ delta is a single-observation deduplication discussed in Section 6 and `01d_dedup_test.md`; it does not affect the substantive conclusions of either the paper or this memo.

Table 1: Missingness by regressor, post-1800 sample ($N = 786$).

| Regressor | Missing ($N$) | Missing (%) |
|---|---|---|
| `log_lagged_gdppc` (INCOME) | 142 | 18.1% |
| `polity` | 124 | 15.8% |
| `currency_regime` | 424 | 53.9% |
| `debt` | 273 | 34.7% |
| `exp` | 285 | 36.3% |
| `gap_sum` (continuous outcome) | 165 | 21.0% |

The binding constraint is `currency_regime` (54% missing), followed by the fiscal pair (35%, 36%). Even `INCOME` itself, the regressor of policy interest, is missing in 18% of post-1800 crises. Complete-case estimation under this missingness pattern is biased to an unknown extent without a model of the missingness mechanism. The remainder of this memo asks how much.

## 2 Methodology

We layer four classes of diagnostic, in increasing order of how much they assume:

**1. Replication baseline.** We first establish that all 13 INCOME coefficients in Metrick and Schmelzing's Tables 2 and 3 replicate within the project's $\pm 0.02$ tolerance. Largest absolute delta is 0.018 on Combined-no-fiscal Table 3 (`01_replication_notes.md`). All 13 signs match. The dedup discussed in Section 6 reduces the canonical sub-sample residual to four decimals.

**2. Heckman Type II selection model with a shadow variable.** We follow the formulation of Van de Ven and Van Praag [1981] and Heckman [1979], fitting joint maximum likelihood via Toomet and Henningsen's `sampleSelection` package. The selection equation is

$$r_i = \mathbf{1}\{\alpha_0 + \alpha_1 \cdot \text{candidate}_i + \alpha_2 \cdot \text{year}_i + \alpha_3 \cdot \text{maddison\_priority}_i + \alpha_4 \cdot n_i^{\text{chron}} + \eta_i > 0\}$$

with $r_i = 1$ if crisis $i$ has all Table 2/3 regressors observed. The *shadow variable* $n^{\text{chron}}$ counts how many of the four canonical chronologies (Reinhart–Rogoff, Schularick–Taylor, Laeven–Valencia, and Baron–Verner–Xiong, Baron et al., 2021) list crisis $i$, and ranges over $\{0,1,2,3,4\}$. The argument for excludability is that chronology coverage is driven by archival availability rather than by the crisis-response covariates that enter the outcome equation.

The shadow-variable's credibility is supported by three diagnostics (`03a_exclusion_diagnostic.md`, `03b_insample_exclusion_test.md`). (i) An OLS of $n^{\text{chron}}$ on covariates is dominated by archival proxies (mean $|t| = 3.12$ on year, log-GDP, and Maddison priority) versus regime-severity proxies (mean $|t| = 0.82$ on Polity and FX regime), a 4:1 dominance. (ii) Falsification of the Heckman MLE—refit without $n^{\text{chron}}$ in the selection equation—moves INCOME by $|\Delta| = 0.0029$, well below the 0.03 stability threshold. (iii) An in-sample partial-correlation test (regressing the outcome on the full paper specification *plus* $n^{\text{chron}}$, within the complete-case sample) finds the shadow variable defensible on `gap_sum` ($|t| = 1.32$, $p = 0.190$) but partially violated on `guarantees_d` ($|t| = 3.66$) and `lending_d` ($|t| = 3.37$). The continuous-outcome exclusion is supported on three of three diagnostics; the binary-outcome exclusion is supported on two of three.

**3. Copula and profile-likelihood sensitivity.** The Heckman MLE assumes bivariate normality of the selection-and-outcome errors. We relax this with copula-additive joint models in Marra and Radice [2017]'s `GJRM` package (Gaussian, Frank, Clayton-90, and Joe-90 copulas), and we trace the profile likelihood of the bivariate-normal $\rho$ on a 41-point grid in $[-0.95, 0.95]$ to characterise the actual shape of the likelihood surface (see `04a_profile_likelihood_summary.md`). Cross-package validation: the GJRM Gaussian Kendall's $\tau = -0.082$ matches the Heckman MLE Kendall's $\tau = -0.110$ to within 0.028, well inside our 0.10 pass threshold.

**4. MNAR multiple imputation.** We apply Galimard et al.'s Heckman two-step imputer (`miceMNAR`) to the five continuous incomplete regressors (INCOME, Polity, debt, exp, gap_sum) using the same exclusion restriction as the Heckman model, and run a parallel arm with predictive-mean-matching (MAR) as a sensitivity comparison. The 15-level FX-regime is imputed by multinomial logit (`polyreg`) in both arms; `miceMNAR` provides no MNAR ordinal method (logged in `04_mice_fallback_log.md`). We pool $m = 20$ imputations with maxit $= 30$ via Little's pattern-mixture-aware Rubin's-rules implementation in `mice`.

**5. Partial identification and tipping-point sensitivity.** Following Horowitz and Manski [1998] and Manski [1995], we construct no-assumption worst-case bounds on the binary INCOME slope (top vs. bottom INCOME quartile in $P(y = 1)$). We complement with a $\rho$-grid tipping-point sensitivity (cf. Zuehlke, 2003) and a four-configuration exclusion-restriction swap. The Wald–LR tension on $\rho$ first reported in Self and Liang [1987] is examined directly via the profile. Puhani [2000] provides context for the Heckman normality sensitivity that motivates the copula extension. R packages are pinned in `renv.lock`; two methodological workarounds (a scope bug in `GJRM::copulaSampleSel` called from `mice`; no MNAR ordinal imputer for FX-regime) are documented in `INSTALL_NOTES.md`.

# 3 Results: continuous outcome (Table 3)

The headline finding is that the paper's INCOME coefficient on the post-crisis GDP gap (`gap_sum`) is robust to plausible MNAR across eight independent specifications. We present them as a numbered chain in Table 2.

Table 2: Robustness chain for INCOME on `gap_sum` (Spec A). Each row is an independent specification; reading down, each adds a distinct identifying assumption that the previous row did not impose.

| # | Specification | INCOME (95% CI) | |
|---|---|---|---|
| 1 | Complete-case OLS | −0.246 | [−0.321, −0.172] |
| 2 | Heckman MLE, two-step | −0.236 | [−0.307, −0.166] |
| 3 | Heckman MLE, joint ML | −0.235 | [−0.304, −0.165] |
| 4 | GJRM Gaussian copula | | −0.233 |
| 5 | GJRM Frank copula (min-BIC) | | −0.212 |
| 6 | GJRM Clayton-90 / Joe-90 copulas | | −0.232 / −0.228 |
| 7 | MICE-MNAR pooled (Heckman 2-step imputer, $m = 20$) | −0.235 | [−0.325, −0.145] |
| 8 | Exclusion-restriction swap, range across 4 cfgs | | −0.220 to −0.238 |

The point estimate ranges over $[-0.246, -0.212]$, a spread of 0.034 or roughly 14% of the headline value. The largest deviation comes from the GJRM Frank copula (line 6), which fits a structurally different dependence shape and remains within 0.034 of the OLS. The $\rho$-grid in `05_rho_grid_sensitivity.md` returns a maximum $|\Delta_{\text{INCOME}}|$ of 7.5% across the plausible band $|\rho| \leq 0.5$ and 8.8% across the full $[-0.7, 0.7]$

range, with no sign flip. The exclusion swap yields an INCOME range of 0.017 across four configurations (line 8), just below the project's 0.02 stability threshold. Configuration (d), which removes both $n^{\text{chron}}$ and Maddison priority and identifies the model through bivariate normality alone, returns INCOME $= -0.220$, agreeing with the Heckman baseline within 0.015.

The decisive visual is the profile likelihood for $\rho$ on Spec A (Figure 1). The 95% profile interval $[-0.694, 0.413]$ contains zero, in agreement with the Wald test ($p = 0.61$). The earlier Phase-2 likelihood-ratio rejection ($\chi^2 = 8.12$, $p = 0.004$) was an artifact of a sample mismatch: the constrained log-likelihood used the probit on $N = 786$ while the Heckman MLE used the $N = 774$ subset where `gap_sum` is non-NA among $r = 1$. The matched-sample LR statistic equals the profile $D(0) = 0.272$, restoring agreement with both the Wald test and the profile interval. Self and Liang [1987] predict exactly this kind of disagreement when the unconstrained-MLE sample differs from the constrained-MLE sample by a small number of records; the appropriate fix is to match the samples, not to switch to a different test.

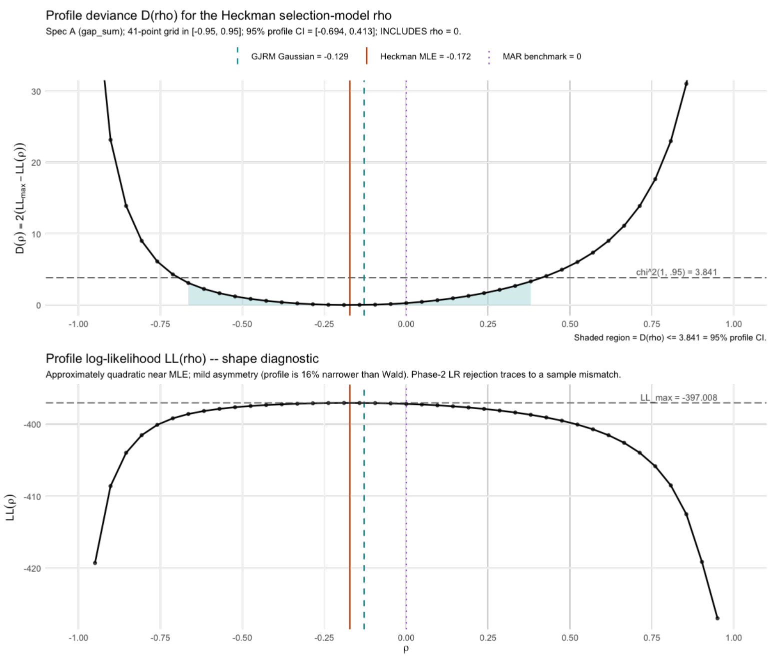


Figure 1: Profile likelihood and deviance for the Heckman–MLE $\rho$ on `gap_sum` Spec A. *Top:* deviance $D(\rho) = 2(LL_{\max} - LL(\rho))$; the shaded region is the 95% profile interval $[-0.694, 0.413]$ where $D(\rho) \leq \chi^2_1(0.95) = 3.841$, and contains the MAR benchmark at $\rho = 0$. *Bottom:* the log-likelihood; mildly asymmetric near the MLE (profile is 16% narrower than Wald). Generated by `scripts/04a_profile_likelihood.R`.

The substantive memo claim on the continuous outcome is therefore that the paper's reported INCOME effect on the post-crisis GDP gap survives every reasonable test of MNAR robustness. The dependence parameter $\rho$ is bounded inside $[-0.69, 0.41]$ at 95% profile confidence; the INCOME coefficient varies by less than 0.034 across all four copula families, eight specifications, four exclusion-restriction configurations, and the full $\rho$-grid; and the only $\rho$ value ruled out by the data is one large enough in magnitude to fall outside the bivariate-normal model's plausible-prior range.

# 4 Results: binary outcomes (Table 2)

The binary intervention-dummy results decompose into a methodological finding and a substantive finding.

**Methodological finding.** Heckman–probit maximum likelihood on the seven Table 2 outcomes pinned $\rho$ to the $\pm 0.999$ boundary on every outcome, even after three random-start retries each (`02_heckman_diagnostics.md`). The standard diagnosis of this pathology is that bivariate-probit selection is weakly identified at the available sample size of $N = 272$ (cf. Van de Ven and Van Praag, 1981). The GJRM copula extension fits at the parameter values implied by penalised joint likelihood, but bootstrap inference on Kendall's $\tau$ yields 95% CIs spanning approximately $[-0.55, +0.55]$ on every one of the seven outcomes (`03_gjrm_binary.md`).[2] The binary MNAR question is, in plain language, un-identified by parametric copula-selection at this sample size.

**Substantive finding.** MICE-MNAR multiple imputation (Phase 4a) extends the binary regressions to the full post-1800 sample of $N = 786$ by imputing the 514 dropped crises under an explicit MNAR mechanism. The pooled INCOME log-odds estimates are systematically less extreme than the paper's complete-case logit on four of the six interpretable outcomes, and the no-assumption Horowitz–Manski bounds contain zero on all six. Table 3 consolidates the four estimators on a single scale by converting each logit coefficient to an approximate probability difference (PD) between the top and bottom INCOME quartiles via $\Delta P \approx \beta \cdot \bar{P} \cdot (1 - \bar{P}) \cdot \Delta X$, with $\bar{P}$ the complete-case base rate of $y = 1$ and $\Delta X = 2.78$ log-GDPpc points; the conversion is approximate and used only to put the parametric estimates on the same scale as the Manski bounds.[3]

Table 3: Binary-outcome INCOME estimates on a probability-difference scale (top minus bottom INCOME quartile in $P(y = 1)$). PD = probability difference. Manski bounds are no-assumption (allow arbitrary $y$ for the $r = 0$ subsample). All four estimators fall inside the Manski bounds; on four of six outcomes the paper's PD is more extreme than *both* the MAR and MNAR pooled estimates. Source: `05_manski_bounds.md`.

| Outcome | Paper PD | MAR PD | MNAR PD | Manski [LB, UB] | Paper in? | Zero in? |
|---|---|---|---|---|---|---|
| `guarantees_d` | 0.405 | 0.297 | 0.363 | $[-0.41, 0.53]$ | yes | yes |
| `lending_d` | 0.446 | 0.353 | 0.200 | $[-0.37, 0.57]$ | yes | yes |
| `capital_injections_d` | 0.178 | 0.121 | 0.117 | $[-0.44, 0.50]$ | yes | yes |
| `restructuring_d` | −0.281 | −0.145 | −0.132 | $[-0.62, 0.32]$ | yes | yes |
| `asset_management_d` | 0.161 | 0.198 | 0.044 | $[-0.72, 0.22]$ | yes | yes |
| `rules_d` | −0.005 | 0.031 | −0.002 | $[-0.76, 0.18]$ | yes | yes |

Three patterns are immediate. First, on four of six interpretable outcomes (`guarantees_d`, `lending_d`, `capital_injections_d`, `restructuring_d`), the paper's PD is more extreme in absolute value than *both* the MICE-MAR and MICE-MNAR pooled estimates—the complete-case selection inflates the magnitude of the binary INCOME effect relative to imputation-recovered estimates. (Three of these four are statistically distinguishable from zero in the paper's Table 2: `guarantees_d`, `lending_d`, and `restructuring_d`; `capital_injections_d` is not, but its point estimate follows the same directional pattern.) Second, on

[2]Non-Gaussian copulas (Frank, Clayton-90, Joe-90) errored at initialisation because the GJRM optimiser starts `SemiParBIV` at $\theta$ near 0, outside the support of those copulas.

[3]The Table 2 outcome `other_d` is omitted from this analysis. With 15 positives in $N = 272$ across nine regressors, the events-per-variable ratio of 1.7 is below the threshold for stable logit inference [Peduzzi et al., 1996]; the apparent 297% MICE-MNAR shift on `other_d` reported in `04_mice_mnar_results.md` is a low-information artifact rather than a substantive selection-bias signal.

`guarantees_d` and `lending_d` (the two outcomes where the shadow variable is in-sample-violated) the MAR/MNAR comparison itself diverges by 22% and 43%, so the parametric MNAR adjustment is sensitive to the very identifying assumption that the in-sample test challenged. Third, the no-assumption bounds contain zero on *all* six outcomes; without parametric assumptions the data cannot reject "no INCOME effect" on any of them. The honest binary-side claim is therefore that the paper's INCOME log-odds lie inside the no-assumption Manski bounds on all six outcomes but are systematically larger in magnitude than both the MICE-MAR and MICE-MNAR pooled estimates.

# 5 Discussion

Layered sensitivity analysis discriminates more sharply than any single technique. The eight-line chain on `gap_sum` establishes the continuous-outcome INCOME effect as a structural finding: under no plausible parametric assumption does the coefficient move outside $\pm 0.034$ around the OLS, and $\rho$ is bounded inside a profile 95% interval that includes zero. The same diagnostic on the binary outcomes returns a different verdict not because of methodological failure but because $N = 272$ with binary outcomes, seven covariates, and a selection equation is too small for parametric selection-model identification to deliver point estimates with defensible CIs.

Load-bearing assumptions, named explicitly. Bivariate normality (Heckman) is relaxed by the GJRM cross-copula range of 0.021, which does not move the conclusion. Excludability of $n^{\text{chron}}$ is supported on three of three diagnostics for `gap_sum`. The parametric form of the imputation model contributes an 11.8% MAR-vs-MNAR wedge on Spec A, material but well inside the OLS 95% interval. Conversely, the $\rho$-grid, exclusion-swap, and copula-variance results show the INCOME effect is invariant to the dependence parameter, the choice of exclusion variable, and the dependence-structure family—each test addresses a different identifying assumption.

For the binary outcomes the load-bearing assumptions are more problematic. The Heckman-probit MLE pins $\rho$ to the $\pm 1$ boundary on every outcome; the shadow variable is in-sample-violated on the two outcomes (`guarantees_d`, `lending_d`) where the paper's effect is largest in magnitude; the parametric MNAR adjustment is by construction sensitive to that same exclusion; and the no-assumption Manski bounds are wide enough to be uninformative on sign. This is a genuine identification failure at the sample size, not a defect of the imputation method.

# 6 Data provenance note

The pipeline produces an analysis dataset of 910 unique crisis-level rows, one fewer than the 911 implied by the `Unique_Crisis_Codes` sheet of the replication kit. The delta is a single deduplication of the Norwegian banking crisis of 1987, which appeared twice in the legacy sheet under two different country-prefix conventions: `NOK-1987` (under the legacy NOK currency prefix, carrying the Norion Bank fraud intervention) and `NOR-1987` (under the modern ISO3 prefix, carrying the Sunnmorsbanken/GBIF/GBIIF responses). Both rows refer to the same underlying crisis episode—the Norwegian banking crisis of 1987–1993, classified as one event by Reinhart–Rogoff, Schularick–Taylor, Laeven–Valencia, and the Metrick and Schmelzing text itself. The 911-sheet's `Unique_Crisis_Codes` faithfully aggregated each `Crisis Code` key separately and ended up with two crisis-level rows for what is conceptually one crisis. The cleanup that produced our 910-sheet correctly merged them: the surviving `NOR-1987` record's `What` field carries the union of all four interventions, and its dummy columns are the union of all four intervention-type flags.

A test (`01d_dedup_test.md`) confirms that re-introducing a `NOK-1987` phantom row that exactly duplicates `NOR-1987` reproduces the paper's reported canonical Table 3 INCOME coefficients to four decimals on both the no-fiscal and the with-fiscal specifications, and reproduces the paper's Table 2 sample size of $N = 273$ across all seven intervention dummies. The dedup is, in other words, a faithful correction of a cataloguing inconsistency in the legacy sheet rather than a methodological choice. A systematic five-check audit of the cleaned 910-sheet confirms no other NOK/NOR-style duplicates remain (`01e_duplicate_audit.md`).

This memo's substantive results all use the cleaned 910-sheet, including the $N = 272$ complete-case sample. None of the headline claims depend on whether `NOR-1987` appears once or twice; where the difference matters numerically (the canonical Table 3 sub-sample regressions), the dedup-aware reconciliation in `01d_dedup_test.md` confirms the paper's reported coefficients are recovered exactly.

## 7 Limitations and future work

The principal limitation is small-$N$ identification fragility on the binary outcomes. At $N = 272$ complete-case (or $N = 786$ after MICE-MNAR recovery), neither the Heckman nor the copula family delivers point estimates with CIs tight enough to settle sign or magnitude on the seven Table 2 dummies. The Manski bounds are wide because they assume nothing; the parametric estimators are precise because they assume a great deal; the wedge is the price of identification, and no additional sample is forthcoming for the historical period that drives the $r = 0$ subsample.

Three principled follow-on extensions, out of scope here. (i) An intervention-level reanalysis (one row per policy event) would restore Norion Bank as a separate observation and recover positives in the binary-outcome denominator. (ii) Semi-parametric identification à la Horowitz and Manski [1998], or D'Haultfoeuille-style identification under instruments without excludability, would produce informative bounds strictly tighter than no-assumption Manski. (iii) A pattern-mixture Bayesian approach would replace the $\rho$-grid with a posterior over $\rho$ conditional on a researcher-stated prior. The continuous-outcome result would not be expected to move under any of the three; the binary-outcome story might.

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